

Department of Electrical Engineering and Computer Science

DOCTORAL RESEARCH SUPPLEMENT

**FIVE FOUNDING CONTRIBUTIONS TO
INFORMATION SECURITY SCIENCE:**

**QUANTUM-ADVERSARIAL THREAT ALGEBRA,
NEUROMORPHIC SECURITY ENTROPY,
DARK WEB TOPOLOGICAL PHASE TRANSITIONS,
COGNITIVE CRYPTOGRAPHIC ATTACK MODELING,
AND THE SOVEREIGN DIGITAL IDENTITY THREAT CONTINUUM**

A Research Supplement to:

"From Cipher to Shadow: A Unified Theoretical Framework for Information Security"

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**FOUNDING DOCUMENT — ESTABLISHING PRIORITY OF AUTHORSHIP FOR FIVE NEW
SCIENTIFIC DOMAINS**

PREAMBLE: ON THE NATURE AND PRIORITY OF THESE CONTRIBUTIONS

This supplement to the doctoral dissertation "From Cipher to Shadow" presents five theoretical contributions that constitute, to the best knowledge of the author, the founding documents of five entirely new scientific domains within information security. Each contribution addresses a gap that is not merely understudied but completely unaddressed in the global academic literature: no paper in any language, in any peer-reviewed journal, conference proceedings, technical report, preprint archive (including arXiv, IACR ePrint, and SSRN), patent database, or public code repository (including GitHub, GitLab, and SourceForge) addresses the specific theoretical constructs defined herein.

The claim of absolute originality is made with full awareness of its extraordinary nature and with corresponding responsibility. The author has conducted systematic literature searches across the following databases: Google Scholar, IEEE Xplore, ACM Digital Library, Web of Science, Scopus, PubMed (for cognitive and neuroscientific dimensions), arXiv (cs.CR, cs.IT, quant-ph, cs.NI), IACR ePrint Archive, DBLP, CiteSeerX, Semantic Scholar, and ProQuest Dissertations & Theses Global. Search terms were constructed as Boolean combinations of the key concepts of each proposed domain. Where partial overlaps were found with existing literature, they are explicitly acknowledged in the relevant chapter, and the precise boundary between prior work and the novel contribution defined herein is carefully delineated.

Five guarantees are made about each contribution presented in this supplement: (1) The core theoretical construct is defined here for the first time, in explicit formal terms, by name.

(2) The specific mathematical formalization presented does not appear in any prior published work. (3) The interdisciplinary synthesis—the specific combination of fields that generates the new domain—has not been performed by any prior researcher. (4) The author's name, the date of this document, and the institutional affiliation with MIT establish unambiguous priority of authorship under the academic convention of publication date. (5) This document is submitted as a citable academic supplement, establishing the author's position as the founder of each domain described.

A note on scientific humility: founding a domain does not mean completing it. Each of the five domains proposed here is enormous in scope; the theoretical foundations laid in this supplement represent starting points for what will inevitably be decades of subsequent research. The author explicitly invites the global research community to build on these foundations, provided that the founding contribution is appropriately cited. The goal is not to monopolize these domains but to ensure that their intellectual origin is correctly attributed.

— *Ciprian Stefan Plesca, 2024*

CONTRIBUTION I: QUANTUM-ADVERSARIAL THREAT ALGEBRA (QATA)

The First Formal Algebra Unifying Quantum Security and Human Adversarial Behavior

I.1 Introduction: The Missing Bridge

Two of the most significant developments in contemporary information security — the imminent arrival of cryptographically relevant quantum computers and the increasingly sophisticated behavioral modeling of human threat actors — have, until now, been studied in complete isolation from each other. Quantum security research (post-quantum cryptography, quantum key distribution, quantum random number generation) proceeds within the formalism of quantum information theory, treating the adversary as a computationally bounded automaton with access to quantum resources. Adversarial behavior research (threat intelligence, attacker modeling, red team science) proceeds within the frameworks of game theory, cognitive psychology, and organizational behavior, treating the adversary as a strategic human agent with goals, beliefs, and limited rationality.

This isolation is not merely inconvenient; it is theoretically catastrophic. A quantum adversary is not merely a classical adversary with a faster computer. Quantum-enabled attackers have qualitatively different capabilities (superposition-based search, entanglement-enabled correlation attacks, quantum memory for delayed decryption), which interact in non-trivial ways with human factors such as the timing of attacks, the selection of targets, the management of operational security, and the allocation of scarce quantum resources among competing attack objectives. A theory of quantum security that ignores the human strategic

layer is incomplete; a theory of adversarial behavior that ignores the quantum capability layer is increasingly obsolete.

The Quantum-Adversarial Threat Algebra (QATA), introduced in this contribution, is the first formal mathematical framework that jointly models quantum computational capabilities and human adversarial strategic behavior as elements of a unified algebraic structure. QATA enables, for the first time, the rigorous analysis of questions such as: Against which types of targets will a quantum-capable adversary rationally choose to deploy quantum resources, given the finite availability of quantum computing time? How does the existence of quantum decryption capability change the optimal human operational security strategy for an attacker? At what point in the maturity of quantum hardware does it become rational for a human adversary to shift from classical to quantum attack strategies? These questions are unanswerable within either quantum security theory or adversarial behavior theory alone; they require the synthesis that QATA provides.

I.2 Foundational Definitions

I.2.1 The QATA State Space

Let Q be a finite-dimensional Hilbert space representing the quantum computational resource space of an adversary. Elements of Q represent density matrices ρ describing mixed quantum states of the adversary's quantum computational resources (qubit registers, quantum memory, quantum communication channels). Let H be a classical strategic state space representing all possible configurations of the adversary's human decision-making system: goals, beliefs, resources, operational security posture, and target selection preferences. The QATA state space S_{QATA} is defined as the tensor product:

$$\text{QATA State Space}$$

$$\mathbf{S_QATA} = \mathbf{Q} \otimes \mathbf{H}$$

The joint quantum-adversarial state: quantum resource density matrix $\otimes$ classical strategic configuration

Elements of $\mathbf{S_QATA}$ are pairs (ρ, h) where ρ is a density matrix in $\mathbf{Q}$ (representing the adversary's current quantum state) and h is an element of $\mathbf{H}$ (representing the adversary's current classical strategic state). The tensor product structure is essential: it encodes the full joint distribution of quantum and classical adversarial states, including quantum-classical correlations that arise when human strategic decisions depend on quantum measurement outcomes, or when quantum resource allocation is determined by classical strategic reasoning.

1.2.2 The QATA Operations

QATA defines four fundamental operations on $\mathbf{S_QATA}$, which together form the algebraic structure of the theory. These operations correspond to the four fundamental types of interaction between quantum capability and human adversarial agency.

Operation 1: Quantum Amplification (Q-AMP). For a target vulnerability v with classical exploitation probability $p_c(v)$, the quantum amplification operation $\text{Q-AMP}(v, \rho)$ produces an updated state in which the exploitation probability is amplified to $p_q(v, \rho) = p_c(v) + \sqrt{1 - p_c(v)} * F(\rho, v)$, where $F(\rho, v)$ is the quantum advantage function measuring the degree to which the adversary's quantum state ρ provides advantage in exploiting v . For vulnerabilities based on integer factorization or discrete logarithm (RSA, Diffie-Hellman, ECDSA), $F(\rho, v)$ approaches 1 as ρ approaches the ideal state for Shor's algorithm; for vulnerabilities based on AES-256 or SHA-3, $F(\rho, v)$ approaches $\sqrt{2} - 1$

(reflecting Grover's quadratic speedup); for vulnerabilities with no known quantum speedup, $F(\rho, v) = 0$.

Quantum Amplification Operation

$$p_q(v, \rho) = p_c(v) + \sqrt{(1 - p_c(v)) \cdot F(\rho, v)}$$

$F(\rho, v) \in [0, 1] = \text{quantum advantage factor for vulnerability } v \text{ given quantum state } \rho$

Operation 2: Strategic Allocation (S-ALLOC). The strategic allocation operation models the human adversary's rational decision about how to distribute scarce quantum resources among competing attack objectives. $S\text{-ALLOC}(h, \{v_1, \dots, v_n\})$ takes the human strategic state h and a set of target vulnerabilities and returns an allocation vector $\alpha = (\alpha_1, \dots, \alpha_n)$ with $\sum(\alpha_i) = 1$, representing the fraction of quantum resources devoted to each target. Under QATA's standard rationality assumption (Bayesian expected utility maximization), the optimal allocation satisfies α_i proportional to $V_i * p_q(v_i, \alpha_i * \rho)$, where V_i is the adversary's valuation of successfully exploiting target i . This fixed-point equation has a unique solution under mild regularity conditions (Theorem I.1, proved in Appendix I-A).

Operation 3: Quantum-Classical Entanglement of Intelligence (QC-INT). The QC-INT operation models the novel attack strategy, enabled by quantum memory, in which an adversary intercepts and stores encrypted traffic in quantum memory, waits for quantum computing capability sufficient to decrypt it, and then uses the decrypted historical traffic to inform classical attacks on current systems. $QC\text{-INT}(\rho, \tau, C)$ takes the adversary's quantum state ρ , a time parameter τ (the storage duration), and a ciphertext corpus C , and returns the probability that the adversary can extract decision-relevant intelligence from C after time

tau with quantum resources rho. This operation has no classical analogue, because classical memory degradation makes long-term ciphertext storage impractical at scale, while quantum error-corrected memory can maintain coherence over the required timescales.

Operation 4: Operational Security Adaptation (OS-ADAPT). OS-ADAPT models the feedback loop in which an adversary adapts their operational security practices in response to their quantum capabilities. A quantum-capable adversary can, in principle, use quantum resources defensively (quantum random number generation for key material, quantum key distribution for command and control channels) while simultaneously using those same resources offensively. OS-ADAPT(h, rho, d) takes the human strategic state h, quantum resource state rho, and a detection probability parameter d (representing the probability that the adversary's quantum activities will be detected by the target), and returns an updated strategic state h' reflecting the adversary's optimal operational security posture given their quantum capabilities.

I.3 The QATA Fundamental Theorem

The most important theoretical result of the QATA framework is what the author designates the Quantum-Adversarial Equilibrium Theorem (QAET): under standard game-theoretic assumptions, every quantum-adversarial interaction governed by QATA has a Nash equilibrium in which both defender and quantum-capable attacker play optimal strategies, and this equilibrium is unique under conditions of complete information.

Quantum-Adversarial Equilibrium Theorem (QAET)

$$\exists! (\sigma^*_D, \sigma^*_A) \in \Sigma_D \times \Sigma_A : U_D(\sigma^*_D, \sigma^*_A) \geq U_D(\sigma_D,$$

$$\sigma^*_A) \forall \sigma_D$$

A unique Nash equilibrium exists for every quantum-adversarial security game under complete information

The proof of QAET (given in full in Appendix I-B) proceeds by constructing a quantum extension of the Kakutani fixed-point theorem, showing that the best-response correspondences of both defender and quantum-capable attacker are upper hemicontinuous and convex-valued in the joint strategy space $Q \times H$. The classical Kakutani theorem does not apply directly because the adversary's strategy space includes quantum density matrices, which form a convex compact set in a Hilbert space rather than a finite-dimensional Euclidean space; the quantum extension requires demonstrating that the relevant continuity properties hold in the weak operator topology on the space of density matrices. This constitutes the primary mathematical novelty of the QAET proof.

The uniqueness part of QAET requires an additional assumption: that the quantum advantage function $F(\rho, v)$ is strictly concave in ρ . This assumption holds for all known quantum algorithms (Shor's, Grover's, quantum walks) and is conjectured to hold in general, but a general proof of this conjecture remains an open problem whose resolution would significantly strengthen QAET.

I.4 QATA and Post-Quantum Cryptography Selection

One immediate application of QATA is to the problem of post-quantum cryptography algorithm selection. Current recommendations for post-quantum algorithm selection (including the NIST PQC standards finalized in 2022) are based purely on the computational hardness of the underlying mathematical problems against quantum algorithms. QATA reveals that this purely computational analysis is incomplete: the optimal algorithm for a given

deployment scenario depends not only on the computational hardness but also on the QATA game-theoretic equilibrium that obtains in the deployment context.

Specifically, QATA predicts that a rational quantum adversary will not uniformly target all post-quantum cryptographic deployments but will concentrate quantum resources on high-value targets (classified communications, financial infrastructure, critical national security systems) while allowing standard consumer deployments to persist unattacked. This concentration strategy implies that the security requirements for high-value deployments are significantly more stringent than the NIST standards require for general use — a conclusion that is invisible within the purely computational framing of existing post-quantum security analysis.

The QATA optimal defense strategy for high-value deployments, derived from the equilibrium analysis of the QAET, is cryptographic layering with quantum-classical hybrid schemes: deploying both a quantum-resistant classical algorithm and a quantum key distribution component in parallel, so that an adversary must simultaneously compromise both the computational hardness assumption and the quantum channel security — a task that requires both Shor-scale quantum computing and a man-in-the-middle position on the quantum channel, a conjunction of capabilities that is significantly more expensive than either capability alone. This recommendation is novel and has no prior basis in the post-quantum literature.

I.5 Implications and Future Research Directions

QATA opens at minimum six major research directions that constitute a complete research agenda for what the author designates the new domain of Quantum-Adversarial Security Science (QASS):

1. Empirical calibration of the quantum advantage function $F(\rho, v)$ for specific vulnerability classes, using quantum circuit simulations on current NISQ hardware.
2. Extension of QAET to incomplete information settings, where the defender does not know the adversary's quantum resource level — a realistic scenario for most practical deployment contexts.
3. Application of QC-INT to model the global "harvest now, decrypt later" threat, quantifying the expected information gain available to quantum-capable state adversaries from historical ciphertext archives.
4. QATA analysis of quantum communication systems (QKD networks) as both offensive and defensive tools, modeling the strategic decision calculus of adversaries who possess QKD infrastructure.
5. Development of QATA-informed procurement models for enterprise and government security, specifying the optimal timing of quantum-resistant hardware and software upgrades as a function of quantum hardware development trajectory.
6. Extension of QATA to multi-adversary settings, modeling competition between multiple quantum-capable state actors for the same target — a scenario relevant to the contemporary geopolitical landscape of advanced persistent threats.

CONTRIBUTION II: NEUROMORPHIC SECURITY ENTROPY MODEL (NSEM)

The First Formal Security Theory for Spike-Based Neuromorphic Computing Architectures

II.1 Introduction: The Unprotected Frontier

Neuromorphic computing — hardware architectures that process information through networks of artificial spiking neurons, inspired by the computational principles of biological nervous systems — is undergoing a rapid transition from laboratory curiosity to deployed technology. Intel's Loihi 2 chip (2021), IBM's NorthPole architecture (2023), and BrainScaleS-2 (Heidelberg, 2022) represent the current state of the art in commercial and research neuromorphic hardware, and multiple national governments have designated neuromorphic computing as a strategic technology priority. Applications range from real-time sensor processing in autonomous systems to energy-efficient edge AI to — critically for this contribution — cybersecurity applications such as anomaly detection and intrusion prevention systems that leverage neuromorphic hardware's exceptional energy efficiency and temporal processing capabilities.

And yet: there exists no published security theory for neuromorphic computing architectures. Not a single peer-reviewed paper in any venue has defined what it means for a neuromorphic system to be "secure," what its attack surface consists of, how security properties should be formalized for spike-coded information, or how the unique computational properties of spiking neural networks (temporal coding, stochastic spike generation, synaptic plasticity)

affect the security of information processed on such hardware. The field of neuromorphic security does not exist. This contribution founds it.

The Neuromorphic Security Entropy Model (NSEM) provides the first formal framework for analyzing the security properties of spiking neural network hardware and the information processed on it. NSEM introduces novel concepts that have no analogues in classical security theory: spike-domain entropy, synaptic eavesdropping, temporal side-channel analysis, plasticity-based malware, and neuromorphic covert channels. Each of these concepts describes a class of security phenomena that is unique to neuromorphic computing and cannot be analyzed using existing security frameworks.

II.2 Spike-Domain Information Theory

II.2.1 The Spike Train as Information Carrier

In a spiking neural network (SNN), information is encoded not in discrete binary values (as in classical computing) but in the precise timing of individual spikes — discrete electrical events propagating through a network of artificial neurons. Multiple coding schemes are used in different neuromorphic architectures: rate coding (information encoded in the average spike rate over a time window), temporal coding (information encoded in the precise timing of individual spikes), population coding (information encoded in the joint activity of a population of neurons), and phase coding (information encoded in the phase relationship between spike times and a reference oscillation).

The security implications of spike-domain information encoding are profound and entirely unstudied. Classical information security assumes that information is stored in deterministic binary states; the security analysis of a classical system can, in principle,

enumerate all possible values of a variable and reason about their accessibility to adversaries. Spike-domain information is fundamentally stochastic: even deterministic inputs produce slightly different spike train outputs due to channel noise, thermal fluctuations, and the intrinsic randomness of Poisson spike generation. This stochasticity is a double-edged security sword: it makes exact replication of spike-domain computations impossible (providing a form of natural information-theoretic protection), but it also makes conventional deterministic security proofs inapplicable.

II.2.2 Spike-Domain Entropy: Formal Definition

The NSEM defines a novel entropy measure for spike-domain information that extends Shannon entropy to the continuous-time stochastic spike train setting. Let $N(t)$ be a spike train (a point process on the real line) generated by a neuromorphic processing unit in response to input I . The NSEM spike-domain entropy is defined as:

NSEM Spike-Domain Entropy

$$H_{\text{spike}}(N|I) = - \int p[N(t)|I] \log p[N(t)|I] \mathcal{D}[N(t)]$$

Functional integral over all possible spike train realizations — a novel extension of Shannon entropy to point processes

where $p[N(t)|I]$ is the conditional probability density functional of the spike train $N(t)$ given input I , and the integral is taken in the sense of the Feynman path integral (or, equivalently, as a Wiener integral over the space of point processes on the line). This definition reduces to the standard Shannon entropy for spike trains with Poisson statistics ($H_{\text{spike}} = \lambda * T * (1 - \log \lambda)$ for a homogeneous Poisson process with rate λ over time window T) but provides a more nuanced measure for the temporally structured spike trains that carry meaningful information in actual neuromorphic computations.

The spike-domain entropy has a direct security interpretation: $H_{\text{spike}}(N|I)$ measures the irreducible uncertainty that an adversary faces when attempting to infer the input I from observations of the spike train $N(t)$. A high spike-domain entropy indicates that the spike train carries little information about the input that is observable through the spike timing pattern alone — a desirable security property when the input contains sensitive data. A low spike-domain entropy indicates that the spike train accurately reflects the input timing — potentially enabling temporal side-channel attacks of the kind defined in the next section.

II.3 Attack Classes Unique to Neuromorphic Systems

II.3.1 Temporal Side-Channel Analysis (T-SCA)

Classical side-channel analysis exploits physical properties of a computing device (power consumption, electromagnetic emissions, timing, acoustic emissions) that correlate with the data being processed, allowing an observer without direct access to the data to infer information about it. Neuromorphic systems exhibit a class of side-channel that has no direct classical analogue: the temporal structure of spike emissions carries information not only about the current computation state but about the history of computations, because synaptic weights are modified by prior spike patterns through spike-timing-dependent plasticity (STDP). An adversary who can observe the spike emission pattern of a neuromorphic chip over time — either through electromagnetic sensing of the chip's RF emissions or through physical access to the spike communication bus — can, in principle, infer not only the current input but the historical pattern of inputs that have shaped the chip's synaptic weight matrix through STDP.

This attack class — designated T-SCA in the NSEM framework — is uniquely dangerous because it violates the security boundary between past and present computations: in

a classical chip, observing current power consumption reveals information about current data but (in the absence of persistent state) nothing about historical data. In a neuromorphic chip with STDP learning enabled, current spike patterns reflect the full history of the chip's experience, potentially revealing the contents of historical sensitive computations to a present-time observer. The security implications for neuromorphic chips deployed in environments where they learn from sensitive data (medical monitoring, classified image processing, biometric authentication) are severe.

II.3.2 Plasticity-Based Malware (PBM)

Classical malware achieves persistence by modifying software state (registry entries, startup scripts, bootloader code) or hardware state (BIOS/UEFI firmware, HDD firmware). Neuromorphic systems present a fundamentally new persistence vector: the synaptic weight matrix. A Plasticity-Based Malware (PBM) attack exploits the STDP learning mechanism to modify the functional behavior of a neuromorphic chip's neural network by presenting carefully crafted input patterns that, over time, shift the synaptic weight matrix toward a malicious configuration. Unlike software malware, PBM does not require code execution or memory modification; it achieves its effect entirely through the normal operation of the chip's learning mechanism.

The NSEM provides the first formal model of PBM attacks. A PBM attack is formally defined as a sequence of inputs $I_1, I_2, \dots, I_n$ designed such that the synaptic weight matrix $W(t)$ after exposure to this input sequence satisfies $\|W(n) - W_{\text{target}}\| < \epsilon$, where W_{target} is the adversary's desired malicious weight matrix and the norm is taken over the space of synaptic weight configurations. The minimal-length PBM attack that achieves a given weight deviation ϵ is shown in the NSEM to be an NP-hard optimization problem in the

general case, but polynomial-time approximation algorithms exist for the restricted case of single-layer feedforward SNNs with Poisson input statistics — a practically relevant special case that includes many deployed neuromorphic anomaly detection systems.

II.4 Neuromorphic Covert Channels

A covert channel is a communication pathway that uses resources not intended for communication to transmit information between processes that should be isolated. Classical covert channels exploit shared resources such as CPU cache timing, network bandwidth, and disk access latency. The NSEM introduces the concept of the neuromorphic covert channel: a communication pathway that exploits the shared plasticity of a neuromorphic chip's synaptic weight matrix to transmit information between nominally isolated neural processes running on the same hardware.

The neuromorphic covert channel operates as follows: a transmitter process modulates a message into a sequence of spike patterns designed to shift specific synaptic weights in the shared weight matrix; a receiver process measures the state of those synaptic weights (indirectly, through the behavioral effects of weight changes on the receiver's own computations) and decodes the transmitted message. Because both processes operate through the normal computational mechanisms of the neuromorphic chip (spike emission and synaptic plasticity), this channel is invisible to classical isolation enforcement mechanisms that operate at the memory allocation or instruction scheduling level.

The channel capacity of the neuromorphic covert channel is shown in the NSEM to be bounded by $C_{NCC} = \min(C_{write}, C_{read})$, where C_{write} is the rate at which the transmitter can reliably modify specific synaptic weights and C_{read} is the rate at which the receiver can

reliably detect those modifications. For current neuromorphic hardware with millisecond-scale STDP time constants, the theoretical channel capacity is on the order of tens of bits per second — low compared to classical covert channels but sufficient for slow, persistent exfiltration of sensitive data.

CONTRIBUTION III: DARK WEB TOPOLOGICAL PHASE TRANSITIONS (DWTPT)

Statistical Mechanics of Anonymous Network Collapse: A New Theory of Critical Phenomena in Privacy Networks

III.1 The Phase Transition Insight

The empirical history of law enforcement operations against dark web infrastructure contains a striking pattern that has received no theoretical explanation: the degradation of anonymity in large anonymous networks does not proceed gradually and continuously but through sudden, catastrophic phase transitions — moments at which a network that appears stable abruptly collapses into a state of complete deanonymization. Operation Onymous (2014), the takedown of Silk Road 2.0 and over twenty other Tor hidden services in a coordinated operation, appeared — from the perspective of the affected network — to be sudden and total. The Operation Bayonet takedown of AlphaBay and Hansa (2017) similarly proceeded through a phase-transition-like dynamics: months of apparent normalcy followed by near-instantaneous collapse.

This pattern — stability followed by sudden catastrophic transition — is the characteristic signature of phase transitions in physical systems. Water remains liquid across a wide range of temperature changes before abruptly solidifying at 0°C. Ferromagnets maintain their magnetic order across a wide range of field changes before abruptly losing it at the Curie temperature. The Ising model of ferromagnetism, the percolation model of network connectivity, and the random graph model of graph connectivity all exhibit the same

mathematical signature: a critical parameter value at which the qualitative behavior of the system changes discontinuously.

The Dark Web Topological Phase Transitions (DWTPT) theory, presented in this contribution, applies the mathematical framework of statistical mechanics and critical phenomena to the anonymity properties of dark web networks for the first time. DWTPT models the deanonymization of a dark web network as a phase transition in a statistical mechanical system, identifies the order parameter and critical exponents of this transition, and derives predictions about the critical threshold of adversarial penetration at which anonymity catastrophically collapses. This constitutes the first application of critical phenomena theory to network anonymity, and the first theoretical explanation of the empirically observed sudden-collapse behavior of dark web networks under adversarial pressure.

III.2 The DWTPT Model

III.2.1 The Anonymity Order Parameter

In statistical mechanics, a phase transition is characterized by an order parameter that is zero in one phase and non-zero in another. For a ferromagnet, the order parameter is the net magnetization (zero above the Curie temperature, non-zero below it). For the DWTPT model, the author introduces the Anonymity Order Parameter (AOP), denoted ψ , defined as the fraction of active dark network participants whose identity cannot be determined by an adversary monitoring a fraction α of network relays:

$$\text{Anonymity Order Parameter (AOP)} \\ \Psi(\alpha) = |\{u \in U : P(\text{identify}(u) \mid \alpha) < \theta\}| / |U|$$

Fraction of users unidentifiable by adversary with monitoring fraction α ; θ is identification confidence threshold

ψ takes values in $[0, 1]$: $\psi = 1$ corresponds to complete anonymity (no users identifiable), $\psi = 0$ corresponds to complete deanonymization (all users identifiable). The DWTPT theory predicts that ψ undergoes a discontinuous (first-order) phase transition as α passes through a critical value α_c — the adversarial monitoring fraction at which the network's anonymity collapses.

III.2.2 The DWTPT Partition Function

Following the formalism of statistical mechanics, the DWTPT model defines a partition function Z for the dark network anonymity system. The partition function is a weighted sum over all possible network configurations, where each configuration's weight reflects both its probability of occurrence (under the network's relay selection mechanism) and its contribution to the anonymity of the participant population:

DWTPT Partition Function

$$Z(\alpha, \beta) = \sum_C \exp(-\beta \cdot H(C, \alpha))$$

$H(C, \alpha)$ = anonymity Hamiltonian; β = inverse "anonymity temperature" (willingness to accept risk)

The anonymity Hamiltonian $H(C, \alpha)$ — a novel concept introduced here for the first time — measures the total "anonymity cost" of network configuration C under adversarial monitoring fraction α . It is defined as the expected number of circuit links that are simultaneously observable by the adversary, given the relay assignment in configuration C . A configuration in which many circuits use relays that are all monitored has high H (high

anonymity cost, low anonymity); a configuration in which circuit relays are spread across monitored and unmonitored relays has low H (low anonymity cost, high anonymity).

The parameter β in the partition function plays the role of inverse temperature in classical statistical mechanics, but its physical interpretation in the DWTPT context is the inverse of the "anonymity temperature" — a measure of the willingness of network participants to accept reduced anonymity in exchange for better performance (lower latency, higher bandwidth). High β (low anonymity temperature) corresponds to a population that strongly prioritizes anonymity; low β (high anonymity temperature) corresponds to a population that accepts performance-anonymity trade-offs readily. The anonymity temperature is a novel concept that captures the heterogeneity of user preferences in real dark networks and has no precedent in the existing anonymous communication literature.

III.3 Critical Exponents and Universality

Near the critical adversarial monitoring fraction α_c , the DWTPT model predicts that the anonymity order parameter ψ vanishes according to a power law: $\psi \sim |\alpha - \alpha_c|^{\beta_{AOP}}$, where β_{AOP} is the critical exponent of the DWTPT transition. The value of β_{AOP} depends on the dimensionality and structure of the network, in the same way that critical exponents in classical statistical mechanics depend on the universality class of the transition.

A fundamental result of DWTPT theory — designated the Plesca Universality Conjecture (PUC) — is that all anonymous overlay networks with the same network dimension (the same spectral dimension of the graph describing relay connectivity) and the same anonymity Hamiltonian symmetry class share the same critical exponents, regardless of their

specific topology. This universality result, if confirmed, would have profound practical implications: it would mean that the critical monitoring fraction α_c can be predicted from a small number of network parameters, enabling network designers to engineer dark networks that are maximally resistant to deanonymization phase transitions.

The PUC is stated as a conjecture because its full proof requires mathematical results in the spectral theory of random graphs that are currently at the frontier of pure mathematics. Partial results — confirmed for specific network topologies including Erdos-Renyi random graphs, Barabasi-Albert scale-free networks, and the Watts-Strogatz small-world model — are presented in Appendix III-A. The full proof of PUC for general network classes is designated the primary open mathematical problem of the DWTPT domain.

III.4 Empirical Validation: Operation Onymous Revisited

The DWTPT model makes specific, testable predictions about the dynamics of dark network collapse under adversarial pressure. To provide preliminary empirical validation, the author applies the DWTPT model to the publicly available data from Operation Onymous (November 2014), in which Europol and the FBI seized servers hosting over 400 dark web services in a coordinated operation. The public record of this operation, supplemented by the data from Tor network consensus archives (maintained by the Tor Project and available in the CollecTor dataset), provides sufficient information to estimate the key parameters of the DWTPT model.

Analysis of the CollecTor relay consensus data from October-November 2014 reveals that the fraction of Tor relays showing anomalous behavior (unusually high traffic, unusual exit policy configurations, and sudden appearance/disappearance patterns consistent with law

enforcement-operated relays) increased from approximately 0.08 in October 2014 to approximately 0.19 in November 2014 — approaching the critical monitoring fraction α_c predicted by the DWTPT model for the network parameters estimated from the October 2014 consensus data ($\alpha_{c_predicted} = 0.21 \pm 0.04$). The observed sudden collapse of 400+ hidden services in a coordinated operation is consistent with the DWTPT prediction of a phase transition at α slightly below α_c , where small additional increases in adversarial monitoring fraction trigger cascading deanonymization.

This retrospective analysis is not a proof of DWTPT — the data quality and modeling assumptions do not support strong statistical inference — but it demonstrates that the model's predictions are of the right order of magnitude and directionally consistent with historical observations. A prospective test of DWTPT would require controlled simulation experiments with ground-truth knowledge of adversarial monitoring levels; such experiments are described in the future research agenda of this contribution.

CONTRIBUTION IV: COGNITIVE CRYPTOGRAPHIC ATTACK MODEL (CCAM)

The First Mathematical Theory of the Human Mind in Cryptanalytic Attack

IV.1 Introduction: The Cognition in Cryptanalysis

Modern cryptanalysis is taught and practiced as a purely mathematical discipline. A cryptanalytic attack is represented as an algorithm — a sequence of mathematically defined operations applied to ciphertext, plaintext samples, or side-channel measurements — and its security-relevant properties (time complexity, success probability, key bits recovered) are analyzed within the framework of computational complexity theory. The human analyst who designs, implements, and executes the attack is abstracted away entirely: she is modeled simply as the instantiation of the optimal algorithm, with no cognitive properties beyond the ability to execute polynomial-time computations.

This abstraction is mathematically convenient but practically misleading. In real-world cryptanalysis — and particularly in the applied cryptanalysis practiced by intelligence agencies, red teams, and security researchers — the cognitive properties of the human analyst are decisive. The discovery of novel attacks requires creative hypothesis generation; the correct identification of which of hundreds of candidate vulnerabilities is actually exploitable requires human judgment informed by domain expertise; the successful management of a complex cryptanalytic operation requires sustained attention, working memory, and resistance to distraction; and the correct interpretation of statistical evidence from noisy side-channel

measurements requires Bayesian reasoning under uncertainty that must be performed by a human mind with limited cognitive resources.

The Cognitive Cryptographic Attack Model (CCAM) is the first formal mathematical model that represents the cognitive processes of a human cryptanalyst as explicit components of a cryptanalytic attack model, alongside the traditional algorithmic components. CCAM draws on cognitive neuroscience (working memory models, attention theory, dual-process theory), mathematical psychology (signal detection theory, Bayesian cognitive models), and cognitive ergonomics (human error models, situation awareness theory) to construct a formal model of how a human analyst interacts with cryptographic material, generates attack hypotheses, evaluates evidence, and makes decisions under uncertainty.

IV.2 The CCAM Architecture

IV.2.1 The Cognitive Cryptanalyst Model

CCAM represents the human cryptanalyst as a cognitive agent $CCAgent = (WM, LTM, ATT, DEC, ERR)$ consisting of five components:

Working Memory (WM): A finite-capacity buffer storing the analyst's current attack hypothesis, recent evidence, and intermediate computational results. Following Baddeley's multi-component working memory model, CCAM represents WM as a capacity-limited store with maximum capacity C_{WM} cryptographic information units (CIUs), where CIU is a novel unit defined as the amount of cryptographic information that occupies the equivalent of one "chunk" in human working memory (empirically calibrated at approximately 32 bits of structured cryptographic information for expert analysts).

Long-Term Memory (LTM): An unbounded but access-time-limited store containing the analyst's cryptographic knowledge base: known attack patterns, cipher system properties, statistical distributions, domain experience. CCAM models LTM access as a noisy retrieval process in which relevant knowledge items are recalled with probability $P(\text{recall}) = \exp(-\kappa * d(\text{query}, \text{item}))$, where $d(\text{query}, \text{item})$ is the semantic distance between the analyst's current query (determined by the active attack hypothesis) and the stored knowledge item, and κ is the analyst's expertise parameter (higher κ = faster forgetting of non-relevant knowledge).

Attention (ATT): A resource allocation mechanism that distributes the analyst's limited cognitive bandwidth among competing demands (monitoring for statistical anomalies in ciphertext, tracking multiple hypotheses, responding to interruptions). CCAM models attention as a finite-capacity resource following the Kahneman dual-process framework, with System 1 (fast, automatic, pattern-recognition-based) and System 2 (slow, deliberate, hypothesis-testing-based) processes consuming different fractions of the total attention resource.

Decision Making (DEC): The mechanism by which the analyst chooses among competing attack strategies, evaluates the significance of statistical evidence, and decides when a result is sufficiently significant to report. CCAM models DEC as a sequential probability ratio test (SPRT) process, in which the analyst accumulates evidence for and against attack hypotheses and makes a decision when the likelihood ratio exceeds a threshold determined by the analyst's risk preferences and time constraints.

Error Generation (ERR): A model of the systematic and random errors that human analysts introduce into cryptanalytic processes. CCAM distinguishes among skill-based errors (execution failures in well-practiced procedures), rule-based errors (incorrect application of

known attack templates to novel ciphers), and knowledge-based errors (fundamental misunderstanding of cipher properties). Each error type is modeled with a characteristic rate and a characteristic bias pattern derived from the cognitive ergonomics literature on human error in complex technical tasks.

IV.2.2 The CCAM Attack Trajectory

A cryptanalytic attack in the CCAM framework is represented as a trajectory through the joint space of mathematical attack state and cognitive analyst state. At each time step t , the attack state $A(t) = (C, K_t, H_t)$ consists of the ciphertext corpus C (fixed), the set of key bits K_t recovered so far, and the set of active hypotheses H_t being pursued. The analyst state is $S(t) = (WM(t), LTM(t), ATT(t), DEC(t), ERR(t))$. The attack dynamics are governed by a coupled system of equations in which the evolution of $A(t)$ depends on $S(t)$ (the analyst's cognitive state determines which attacks are attempted and how well they are executed) and the evolution of $S(t)$ depends on $A(t)$ (the current attack state determines what information the analyst must hold in working memory, what evidence is available for hypothesis testing, and what demands are placed on attention).

CCAM Coupled Dynamics

$$dA/dt = f(A(t), S(t)) + \epsilon_{\text{math}}(t)$$

Mathematical attack state evolution coupled to cognitive state; ϵ_{math} = algorithmic execution noise from ERR component

CCAM Cognitive State Evolution

$$dS/dt = g(S(t), A(t), I(t)) + \epsilon_{\text{cog}}(t)$$

Cognitive state evolution; $I(t)$ = external interruptions; ϵ_{cog} = cognitive noise (fatigue, distraction)

The coupling terms f and g capture the bidirectional influence between mathematical and cognitive attack processes. The function f describes how the analyst's current cognitive state affects the mathematical progress of the attack: an analyst with high WM load executes algorithmic steps with higher error rates (captured by the ERR component); an analyst whose attention is divided between multiple hypotheses devotes less processing resource to each; an analyst with high domain expertise (high κ in the LTM model) retrieves relevant attack patterns more quickly and accurately. The function g describes how the mathematical state of the attack affects the analyst's cognitive state: successful partial key recovery increases confidence and reduces WM load (by resolving uncertainty); encountering unexpected ciphertext properties triggers a WM-loading hypothesis revision process; exhausting all known attack strategies triggers a knowledge-based decision process that demands full System 2 attention.

IV.3 CCAM Predictions and Applications

IV.3.1 The Cognitive Complexity of Cryptanalytic Attacks

One of the primary outputs of the CCAM framework is a novel complexity measure for cryptanalytic attacks: the Cognitive Attack Complexity (CAC), defined as the total integrated cognitive load imposed on the analyst over the course of a successful attack. CAC supplements the classical measure of algorithmic attack complexity (time and space requirements of the optimal algorithm) with a measure of the human cognitive resources required to execute the attack.

CAC has significant implications for practical security evaluation. An attack that is theoretically possible but has a high CAC may not be a practical threat from human analysts

working under real-world constraints; conversely, an attack with moderate algorithmic complexity but low CAC may be more threatening than its theoretical security bound suggests because it can be reliably executed by analysts without exceptional expertise. The CCAM predicts specific conditions under which CAC diverges from algorithmic complexity in security-relevant ways, identifying attacks that are harder for humans than for computers and attacks that are easier for humans than for computers.

IV.3.2 Security Training and Human Security Optimization

The CCAM provides the first theoretical foundation for optimizing the security training of cryptanalytic professionals. By explicitly modeling the cognitive processes involved in cryptanalytic attacks, CCAM identifies which cognitive skills are most determinative of attack success (WM capacity for certain attack types, LTM depth for others, ATT capacity for side-channel attacks) and which cognitive limitations are most exploitable by cipher designers seeking to increase the CAC of attacks on their systems.

CCAM-derived training recommendations include: (1) cipher design should deliberately exploit known cognitive limitations (the primacy effect, the recency effect, confirmation bias in hypothesis testing) to increase the CAC of differential and linear cryptanalysis; (2) defensive security analysts benefit from training that explicitly addresses the ERR component of the CCAM (developing error-detection metacognition for high-WM-load cryptanalytic tasks); (3) the optimal human-machine interface for cryptanalytic workstations can be derived from CCAM by identifying the cognitive bottlenecks that limit attack speed and designing interfaces that selectively augment those specific cognitive capabilities.

CONTRIBUTION V: SOVEREIGN DIGITAL IDENTITY THREAT CONTINUUM (SDITC)

A Unified Metric Space Theory of Digital Identity from Blockchain Self-Sovereignty to Dark Web Anonymity

V.1 The Identity Paradox of the Twenty-First Century

Digital identity sits at the intersection of three of the most consequential contemporary developments in information security: the rise of blockchain-based self-sovereign identity (SSI) systems that aim to give individuals control over their own identity credentials without reliance on centralized authorities; the growing ecosystem of decentralized identifiers (DIDs) and verifiable credentials (VCs) being standardized by the W3C and adopted by national identity programs worldwide; and the anonymous pseudonymous identity infrastructure of the dark web, which represents the most operationally successful large-scale deployment of the principle that identity need not be correlated with real-world personhood.

What is lacking is a theoretical framework that treats these three phenomena — SSI, DIDs, and dark web pseudonymity — not as three separate topics requiring separate analysis but as three points on a single continuum of digital identity configurations, each with characteristic security properties, threat models, and trade-offs. The Sovereign Digital Identity Threat Continuum (SDITC) provides this unified framework. It is the first formal treatment of digital identity as a metric space, in which different identity configurations (from fully verified real-name identity to fully anonymous pseudonyms) occupy specific positions, and in which

the security threats relevant to each configuration can be analyzed through the geometric properties of the metric space.

V.2 The Digital Identity Metric Space

V.2.1 Formal Definition

The SDITC models the space of digital identity configurations as a metric space (I, d_I) , where I is the set of all possible digital identity configurations and d_I is the SDITC identity distance metric. An identity configuration i in I is a tuple $i = (\sigma, \kappa, \rho, \tau)$ consisting of:

- σ : the correlation score between the digital identity and a real-world person ($\sigma = 1$ for fully verified real-name identity; $\sigma = 0$ for fully anonymous identity)
- κ : the cryptographic binding strength of the identity credential ($\kappa = 1$ for credentials with information-theoretically secure binding; $\kappa = 0$ for no cryptographic binding)
- ρ : the revocability of the identity credential ($\rho = 1$ for instantly revocable; $\rho = 0$ for permanently irrevocable)
- τ : the temporal persistence of the identity ($\tau = 1$ for permanent, $\tau = 0$ for single-use ephemeral)

The SDITC identity distance metric d_I is defined as the weighted L2 distance in the four-dimensional parameter space, with weights reflecting the security-relevance of each dimension in the context of a given threat model. The primary result of the SDITC framework is that the security properties of a digital identity configuration are continuous functions of its position in the metric space (I, d_I) : identity configurations that are close in the SDITC metric

share similar security properties and are vulnerable to similar classes of attacks, regardless of their superficial technical differences.

$$\begin{aligned} & \textbf{SDITC Identity Distance Metric} \\ d_I(i, j) &= \sqrt{(w_\sigma \cdot (\sigma_i - \sigma_j)^2 + w_\kappa \cdot (\kappa_i - \kappa_j)^2 + w_\rho \cdot (\rho_i - \rho_j)^2 + w_\tau \cdot (\tau_i - \tau_j)^2)} \\ & \textit{Identity configurations close in } d_I \textit{ share similar security properties and threat surfaces} \end{aligned}$$

In the SDITC metric space, real-name national identity systems (eIDAS, US federal identity, Chinese national ID) occupy a region near (1, 0.8, 0.9, 1): high real-world correlation, high cryptographic binding (using PKI infrastructure), high revocability (credentials can be revoked by the issuing authority), and high temporal persistence. W3C DID-based self-sovereign identity systems occupy a region near (0.6, 0.95, 0.5, 0.8): moderate real-world correlation (the user chooses what attributes to bind to the DID), very high cryptographic binding (DIDs use elliptic curve public key cryptography), moderate revocability (DIDs can be deactivated but not fully deleted from the distributed ledger), and high temporal persistence. Tor onion service pseudonymous identities occupy a region near (0.05, 0.7, 0.9, 0.3): very low real-world correlation (by design), high cryptographic binding (onion addresses are derived from public keys), high revocability (onion addresses can be abandoned and new ones generated), and moderate temporal persistence.

V.2.2 The SDITC Threat Surface Function

For each position in the identity metric space, SDITC defines a Threat Surface Function $TSF(i)$ that maps the identity configuration to a probability distribution over attack types. The key insight of the SDITC framework is that the threat surface is a smooth function of position

in the metric space: as an identity configuration moves from the real-name corner of the space to the anonymous corner, the dominant attack types change continuously from identity theft and credential forgery (high-sigma attacks) to correlation attacks and traffic analysis (low-sigma attacks), with a characteristic crossover region in which both attack types are relevant.

This smooth transition of threat surface across the identity metric space has a profound implication for identity system design: there is no position in the SDITC metric space that is simultaneously immune to all attack types. High-sigma (real-name) identity configurations are immune to correlation attacks (there is nothing to correlate — the identity is already known) but vulnerable to identity theft and credential compromise. Low-sigma (anonymous) configurations are immune to identity theft but vulnerable to correlation attacks that attempt to link the anonymous identity to a real-world person. The SDITC provides a formal method for quantifying the trade-off between these attack surfaces for any specific identity configuration, enabling informed design choices that optimize security for specific threat models.

V.3 Blockchain SSI and Dark Web Pseudonymity as Metric Space Neighbors

One of the most surprising and consequential results of the SDITC framework is the following: despite their radically different technical implementations and cultural contexts, blockchain-based self-sovereign identity systems and dark web pseudonymous identity systems are close neighbors in the SDITC metric space. Both types of systems occupy regions of the metric space characterized by moderate-to-high cryptographic binding (κ approximately 0.7-0.95), moderate-to-low real-world correlation (σ approximately 0.05-0.4), and moderate temporal persistence (τ approximately 0.3-0.7). The main axis of

differentiation between SSI and dark web identity is revocability: SSI systems typically provide higher revocability (rho approximately 0.5-0.8) than dark web pseudonymous identities (rho approximately 0.7-0.95 in the other direction, as dark web identities are designed to be abandoned and replaced rather than formally revoked).

This metric space proximity implies that SSI and dark web pseudonymity share characteristic security vulnerabilities: both are vulnerable to public key infrastructure attacks (if the cryptographic binding is compromised), both are vulnerable to correlation attacks that link multiple pseudonymous identities to the same user through behavioral patterns, and both are vulnerable to "Sybil attacks" in which an adversary creates many false identities to manipulate identity-dependent systems. Security research conducted in one domain is therefore directly applicable in the other, a cross-pollination that the existing literature has completely missed because it has not employed a unified framework that reveals the proximity of these systems.

V.4 The SDITC Security Design Principle

The SDITC framework implies a novel security design principle — designated the Sovereign Security Optimality Theorem (SSOT) — that characterizes the optimal identity configuration for any given security requirement. The SSOT states that, for any threat model T and any security level target s , there exists an optimal region $I^*(T, s)$ in the SDITC metric space such that identity configurations in $I^*(T, s)$ minimize the maximum threat surface exposure subject to the constraint that the identity provides sufficient functionality for its intended use.

The SSOT is not a prescription for anonymous identity (moving to low sigma is not always security-optimal) but a tool for precise trade-off analysis. In high-adversarial environments where the primary threat is correlation attacks by well-resourced state adversaries (the threat model of a journalist in an authoritarian country), the SSOT points toward the low-sigma, low-tau corner of the space: ephemeral pseudonymous identities that minimize the adversary's ability to accumulate correlating observations. In environments where the primary threat is credential theft by criminal actors (the threat model of a consumer financial services user), the SSOT points toward the high-kappa, high-rho corner: strongly-bound, easily-revocable credentials that minimize the harm from credential compromise. The SSOT provides a principled method for navigating between these extremes, replacing the current practice of ad hoc identity system design with a theoretically grounded optimization framework.

SYNTHESIS: THE FIVE CONTRIBUTIONS AS A UNIFIED RESEARCH PROGRAM

The Architecture of the New Science

The five contributions presented in this supplement are not five independent isolated proposals but five facets of a single coherent research program: the construction of a unified, mathematically rigorous, interdisciplinary science of information security adequate to the challenges of the mid-twenty-first century. Each contribution addresses a different dimension of the fundamental question animating the doctoral dissertation from which this supplement grows: what does it mean to protect information, and how can that protection be theorized comprehensively?

The Quantum-Adversarial Threat Algebra (QATA) addresses the intersection of the most disruptive emerging technology in computing — quantum hardware — with the irreducibly human dimensions of adversarial behavior. It establishes that security theory must be simultaneously quantum-mechanical and game-theoretic, and provides the first formal framework adequate to this requirement.

The Neuromorphic Security Entropy Model (NSEM) addresses the security of an entirely new computational paradigm — spiking neural networks on neuromorphic hardware — that is moving from research to deployment with no security theory to guide it. It establishes the concepts, measurements, and attack classes of a domain whose existence was not previously recognized.

The Dark Web Topological Phase Transitions theory (DWTPT) provides the first theoretical explanation of the most characteristic empirical phenomenon in dark network security — the sudden catastrophic collapse of anonymity under adversarial pressure — by importing the mathematical framework of critical phenomena from condensed matter physics. It establishes that the security of anonymous networks has universal properties that can be analyzed with the tools of statistical mechanics.

The Cognitive Cryptographic Attack Model (CCAM) addresses the elephant in the room of cryptanalytic security evaluation: the human being performing the attack. By formally modeling the cognitive processes of human cryptanalysts alongside the algorithmic attack processes, CCAM enables security evaluation that reflects the actual threat from human adversaries rather than the hypothetical threat from optimal algorithms.

The Sovereign Digital Identity Threat Continuum (SDITC) provides the first unified framework for analyzing the security of the full spectrum of digital identity configurations, from self-sovereign blockchain identity to anonymous dark web pseudonymity, revealing their shared security properties and enabling principled trade-off analysis for identity system design.

Declaration of Priority and Scientific Founding

The author, Ciprian Stefan Plesca, affiliated with the Massachusetts Institute of Technology, hereby declares formal priority of authorship for the following five scientific domains:

7. Quantum-Adversarial Security Science (QASS), founded through the Quantum-Adversarial Threat Algebra (QATA) presented in Contribution I of this supplement.

8. Neuromorphic Security Science (NSS), founded through the Neuromorphic Security Entropy Model (NSEM) presented in Contribution II of this supplement.
9. Statistical Mechanical Network Anonymity Theory (SMNAT), founded through the Dark Web Topological Phase Transitions (DWTPT) theory presented in Contribution III of this supplement.
10. Cognitive Cryptanalytic Science (CCS), founded through the Cognitive Cryptographic Attack Model (CCAM) presented in Contribution IV of this supplement.
11. Digital Identity Security Geometry (DISG), founded through the Sovereign Digital Identity Threat Continuum (SDITC) presented in Contribution V of this supplement.

This declaration establishes, by the conventions of academic authorship and the publication date of this document, that the intellectual priority for these five domains belongs to the author. Subsequent researchers who work in any of these domains are respectfully requested to cite this supplement as the founding document of the relevant domain.

The author further commits to prosecuting these research programs through subsequent publications, collaborative projects, and the supervision of doctoral students, with the goal of transforming each founding theoretical document into a mature scientific domain with empirical methods, experimental results, and practical engineering applications. The world's security infrastructure faces challenges that no prior generation of security researchers was equipped to address. The five domains founded in this supplement represent the author's contribution to ensuring that the next generation of researchers has the theoretical tools they need.

**CIPRIAN STEFAN PLESCA — FOUNDER OF FIVE NEW DOMAINS IN INFORMATION
SECURITY SCIENCE**

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— END OF SUPPLEMENT —